Cheng Yuan

Contact Information

Fudan University

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Research

Partial differential equations in physics and finance. Mainly focus on: Boundary Layer theory; Regularity theory on degenerate parabolic equation.

Education

School of Mathematical Sciences, Fudan University

Ph.D. student, Mathematics, 2022 - 2027

- Advisor: Prof. Zhen Lei
- GPA in Specialized Courses: 3.78/4.0 (Ranking: 2/100)
- Relevant Courses: Modern PDEs, Modern ODEs, Nonlinear Wave Equations, Functional Analysis, Harmonic Analysis, Probability Theory

School of Mathematics and Statistics, Chongqing University

B.S. in Mathematics, 2018 - 2022. GPA in Specialized Courses: 4.0/4.0 (Ranking: 1/160)

Publications

Preprints

[6] *A Good-Vorticity Low-Regularity Integrator for the Inviscid Oldroyd–B System with Stress Diffusion*, with X. Cheng, Z. Luo and Z. Yang, **Preprint** arXiv: 2610.41250. (Submitted)

[5] *On Cauchy problems for parabolic equations with rough coefficients*, **Preprint** arXiv: 2608.29984. (Submitted)

[4] *Global well-posedness and regularity of the dynamical Prandtl equation*, with H. Jia and Z. Lei, **Preprint** arXiv: 2606.13134. (Submitted)

[3] *Global well-posedness and uniform-in-time vanishing damping limit for the inviscid Oldroyd-B model*, with X. Cheng, Z. Luo and Z. Yang, **Preprint** arXiv:2410.09340. (Submitted)

Journal Articles

[2] *Sharp asymptotic stability of Blasius profile in the steady Prandtl equation*, with H. Jia and Z. Lei, Advances in Mathematics, Volume 480, Paper No. 110533, 65, 2025.

[1] *Backward Uniqueness for 3D Navier-Stokes Equations With Non-Trivial Final Data and Applications*, with Z. Lei and Z. Yang, International Mathematics Research Notices, Volume 2024, Issue 20, Pages 13417-13431.

Visiting

Visiting Scholar, Chinese Academy of Sciences, Sept.-Oct., 2026.

Visiting Scholar, University of Minnesota, March-May, 2025.

Conference & Seminar Talks

PDE Seminar, CAS (AMSS), October 2026. **Speaker**

Conference on Mathematics Fluid Dynamics, Feishu Shanghai, July 2026.

The 11th Shanghai International Symposium on Nonlinear Sciences and Applications, Guangxi Normal University, July 2026. **Speaker** & **Session hoster**

The 15th AIMS: Special Session 81, Athens, July 2026. **Speaker**

2nd Frontiers in Fluid and Kinetic PDEs, CAS (AMSS), July 2026. **Speaker**

PDE Seminar, CAS (AMSS), June 2026. **Speaker**

Conference on Energy Cascade, Singular limits, Dynamical problems in PDEs, Fudan University, April 2026.

SYSU 5th Anniversary Conference on PDEs, Dynamical Systems and Computational Mathematics, SYSU (Shenzhen), March 2026. **Speaker**

Academic Meeting of the NSFC Major Program âFundamental Theory and Applications of Fluid Mechanics.â, Xinyang (XYNU), December 2025. **Speaker**

The 33rd Yangtze River PDE Conference, Nanjing (NUAA), December 2025. **Speaker**

Inaugural "Jiao Chuang" National High-Level Doctoral Academic Forum, Shanghai Jiao Tong University, November 2025. **Speaker**

The 3rd National Doctoral Academic Forum on Mathematical Frontiers, Peking University, October 2025. **Speaker**

23nd Annual Meeting of CSIAM, Changsha, October 2025.

Advances in Mathematical Fluid Dynamics Conference, Peking University, June 2025.

Advanced PDE Seminars, Zhejiang University, June 2025.

Rivière-Fabes Symposium, Minnesota University, April 2025.

Fluid Seminar, Minnesota University, April 2025. **Speaker**

Applied Math Phd Seminar, Fudan University, February 2025. **Speaker**

PDE Seminar, Duke Kunshan University, January 2025. **Speaker**

Workshop on Differential Equations and Nonlinear Analysis, Fudan University, December 2024. **Organizer**

Fluid PDE Seminar, Fudan University, November 2024. **Speaker**

Workshop on Mixing, Enhanced Dissipation and Stability Effects in Fluid Dynamics , Tianyuan Mathematics Research Center, November 2024.

The 2nd National Doctoral Academic Forum on Mathematical Frontiers, Peking University, October 2024. **Speaker**

22nd Annual Meeting of CSIAM: Special Session on Mathematical Theory of Fluid Mechanics , Nanjing, October 2024. **Speaker**

2023 Annual Meeting of CMS, Dalian, December 2023.

1st Workshops on Mathematical Fluid Dynamics Dynamic system, Westlake University, October 2023. **Speaker**

Workshop on Quantitative Uniqueness Theory, Fudan University, June 2023.

Teaching

Fall 2023 Teaching Assistant, Equations of Mathematical Physics Partial Differential Equations (Honors Course)

Grants

2024.05–2026.12 NSFC 123B2008 (300,000 CNY), *Theory of Boundary Layer in Fluid.*

Honors & Awards

2025	The 17th session Academic Star, Fudan University
2025	Outstanding Student Model, Fudan University
2023, 2025, 2026	National Prize Scholarship for PhD Students, Fudan University
2022	XiangHui Scholarship for PhD Students, Fudan University
2022	Outstanding Graduates, Chongqing City
2021	First Prize in Chinese Mathematical Competitions for University Students, CMS
2020	Outstanding Student Model, Chongqing University
2020	National Prize Scholarship for Undergraduate Students, Chongqing University
2020	First Prize in Chinese Mathematical Competitions for University Students, CMS

Services

Conference and Workshop Organization

Workshop on Differential Equations and Nonlinear Analysis, Fudan University, December 2024. (Organization with X.Cheng and Z.Luo)

Seminar and Lecture Series Organization

2020-2022 Organizer and speaker of weekly lectures for CMC in Chongqing University.
2024-now Organizer of the weekly PDEs Seminar in Fudan University.

Languages

Chinese (Native), English (Fluent)